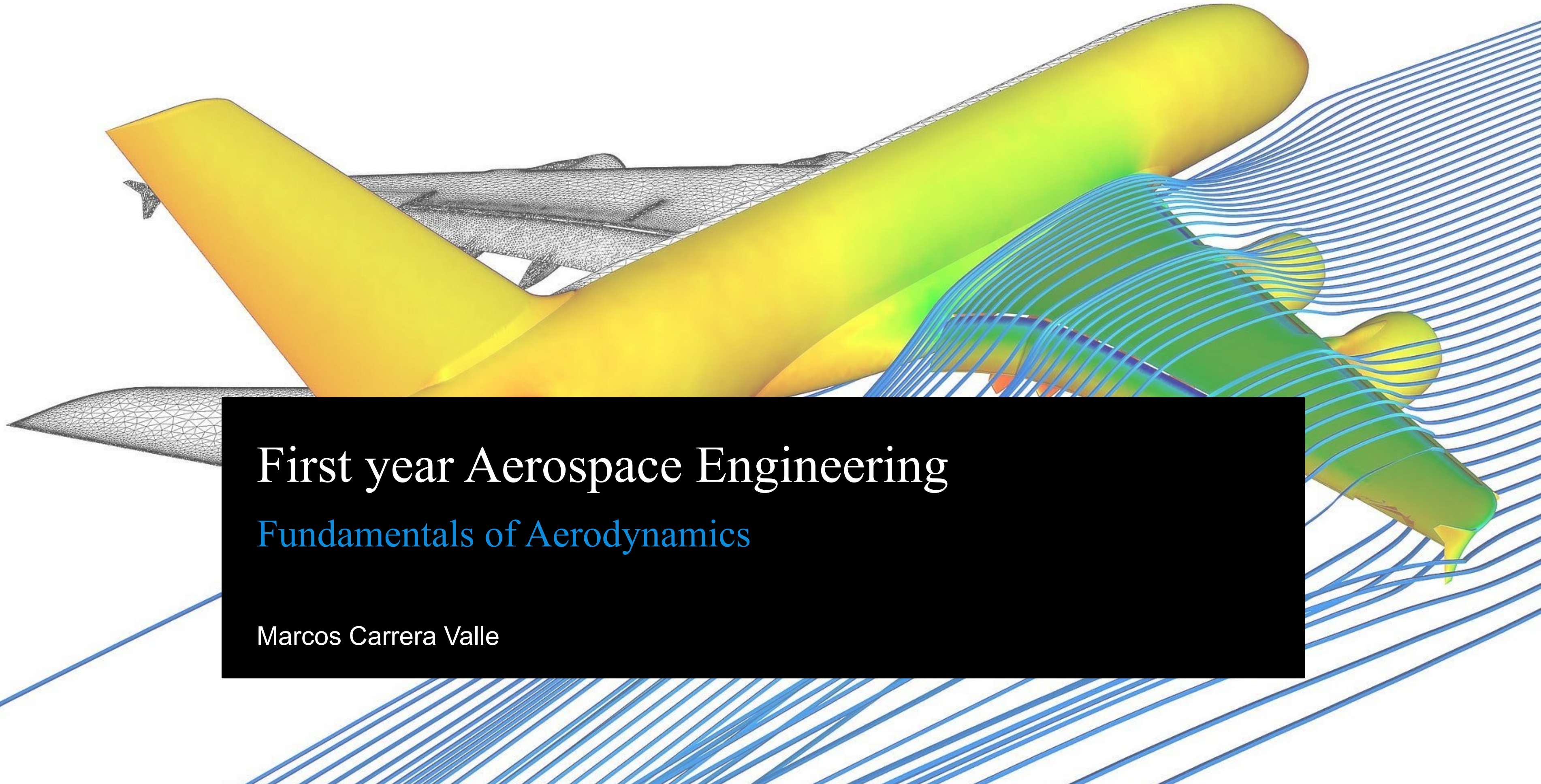


First year Aerospace Engineering

Fundamentals of Aerodynamics

Marcos Carrera Valle

Fundamentals

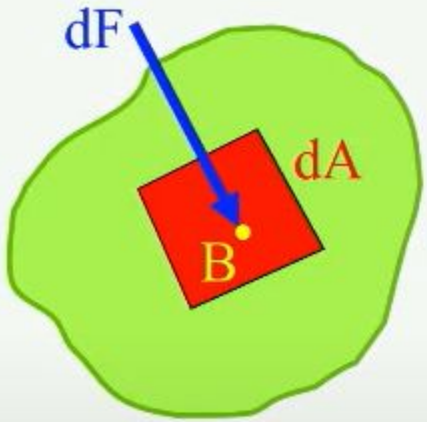
First we will look at the basics:

- Fundamental Quantities
  - Pressure, p (Pa)
  - Density, ρ (kg/m^3)
  - Temperature, T (K)
  - Velocity, V (m/s)

- Fundamental Equations
  - Equation of state
  - Continuity
  - Euler
  - Bernoulli

Fundamental Quantities

Pressure - **Normal force** per unit area on a surface
Measure in Pa (Pascal) or Atm (1 standard atmosphere = 101,325 Pa)



$$p = \lim_{dA \to 0} \left(\frac{dF}{dA} \right) \left[\frac{N}{m^2} \right]$$

Fundamental Quantities

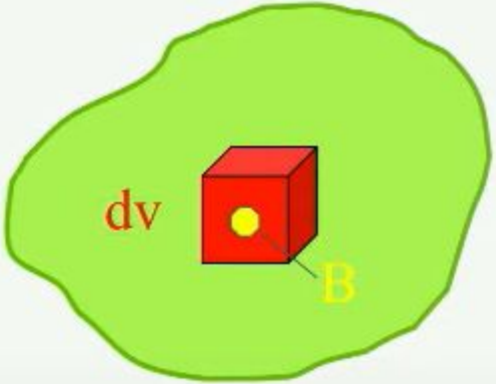
A can of coke is pressurized. Inside the can, the pressure is 3 atm.
The can is at **sea level conditions**, where pressure is 1 atm.
The can is a perfect cylinder with radius 3 cm.

What is the pressure force exerted from the **interior** of the can on the top surface of the can?

What is the **resultant** force acting on the top surface of the can?

Fundamental Quantities

Density - Mass per unit volume
How heavy 1 m^3 of a substance is



The density in point B is:

$$\rho = \lim_{dv \to 0} \left(\frac{dm}{dv} \right)$$

Fundamental Quantities

The can is a perfect cylinder with radius 3 cm, and height 12 cm.
The coke inside weighs 350 grams.

What is the **density** of coke?

Fundamental Quantities

Temperature - Average kinetic energy of a particle in a gas
How hot a substance is

$$KE = \frac{3}{2}kT$$

k is the Boltzmann constant: $k = 1.38 \cdot 10^{-23}\,\frac{J}{K}$

Dimension of temperature: K (Kelvin) or °C (degrees Celsius)

Conversion: $T(K) = 273.15 + T(^{\circ}C)$

Fundamental Quantities

I want to make sure the coke is cold before I drink it.
Currently it is 15ºC, but I want to lower it to 277.85 K.

What is the temperature drop needed?

How much kinetic energy will be lost?

Standard conditions

$$p_s = 1.01325 \cdot 10^5 \frac{N}{m^2}$$

$$\rho_s = 1.225 \frac{kg}{m^3}$$

$$T_s = 288.15K$$

Equation of state

Luckily for us, there is an equation that relates density, pressure and temperature for **perfect gases** - the **equation of state**

$$\boxed{p = \rho R T}$$

R is the **gas constant**:

$$\boxed{R = 287.0 \frac{J}{kg \cdot K}}$$

Fundamental Quantities

The pressure in Amsterdam is standard sea level, and it is currently very cold, only 3ºC.

What is the air density?

It is also 3ºC in base camp of Mt. Everest, however the pressure is much lower here, only ⅓ of standard sea level.

What is the air density of Mt. Everest?

Specific volume

Specific volume is the **inverse** of density

Density is how heavy 1 m^3 of a substance is (kg/m^3)

Specific volume is the volume of 1 kg of a substance (m^3/kg)

$$\nu = \frac{1}{\rho}$$

The equation of state then becomes:

$$p\nu = RT$$

Specific Volume

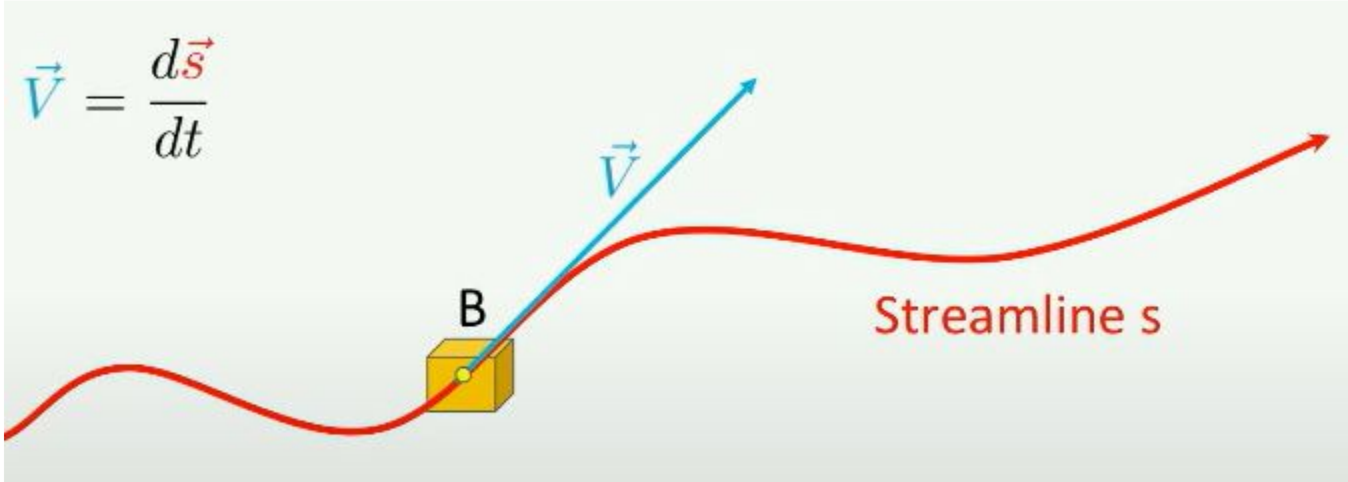
The can is a perfect cylinder with radius 3 cm, and height 12 cm.
The coke inside weighs 350 grams.

What is the **density** of coke?

What is the **specific volume** of coke?

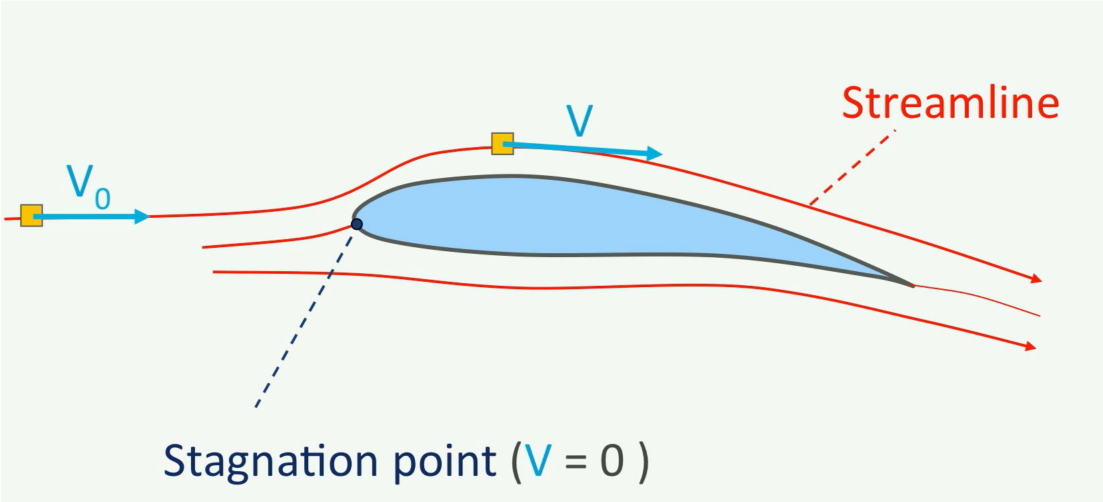
Velocity

How fast something is going (m/s)
For an infinitesimally small particle **B**:

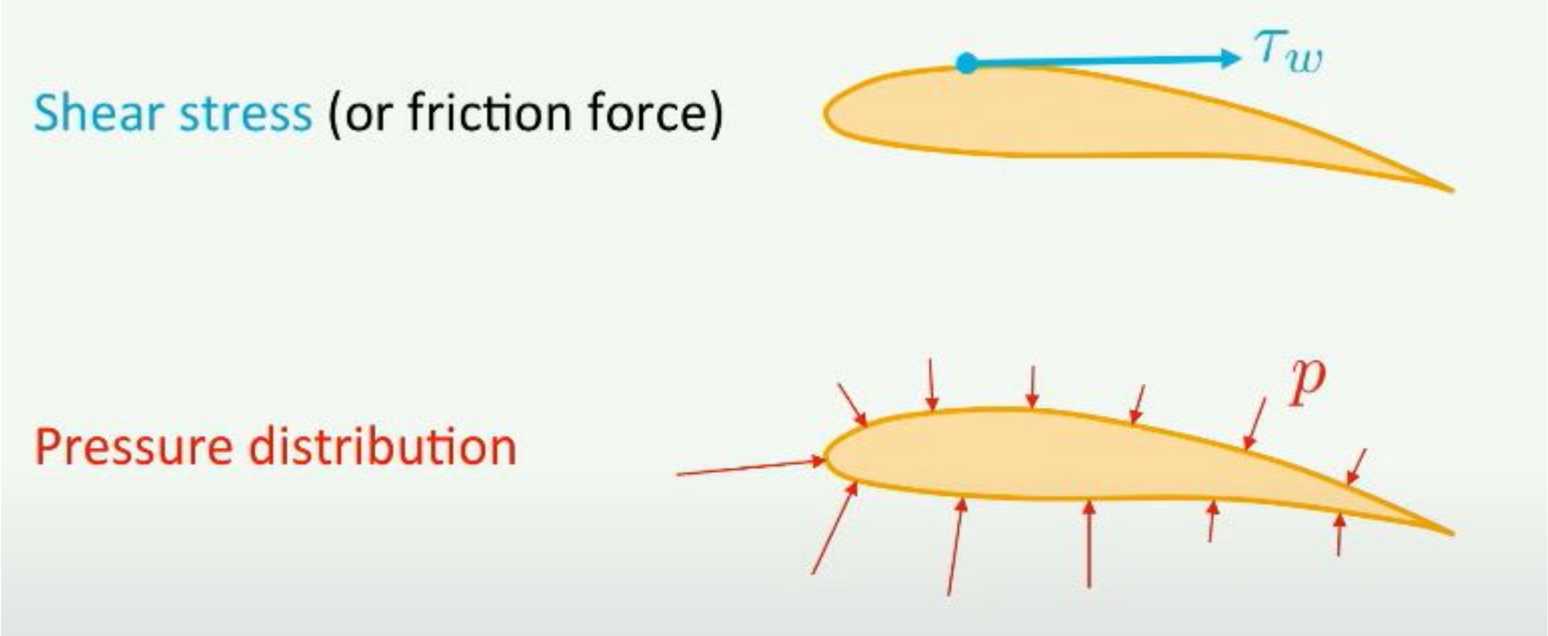


Velocity

At the stagnation point, pressure = **total pressure p_0**



Aerodynamic forces



Fundamental Equations

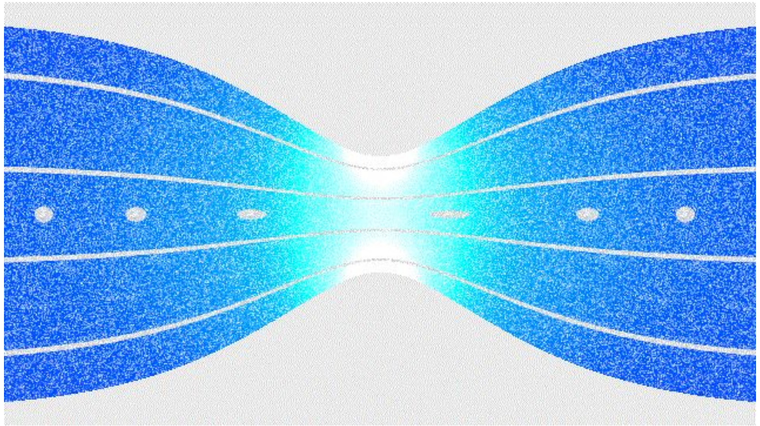
- Conservation of mass
    - Continuity Equation

- Conservation of momentum
    - Euler
    - Bernoulli

Fundamental Equations

Continuity equation

Mass in = Mass out (Mass is conserved)



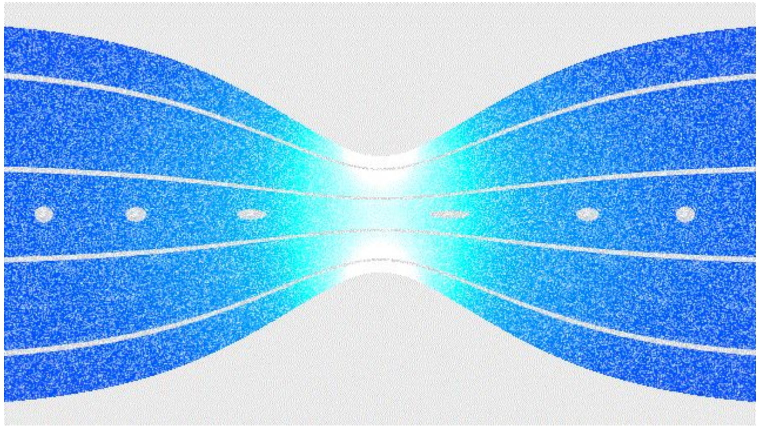
Fundamental Equations

Continuity equation

$$\boxed{\begin{aligned}\rho A V &= \text{constant}\\ \rho_1 A_1 V_1 &= \rho_2 A_2 V_2\end{aligned}}$$

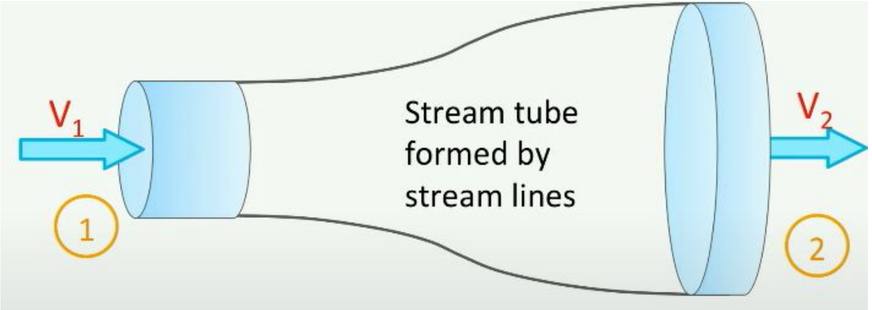
Continuity equation for **steady** flow.

Assumption: ρ and V are *mean* values.



For incompressible flow: $\boxed{A \cdot V = \text{constant}}$

Fundamental Equations



Continuity equation

In the image you can see water (ρ = 1000 kg/m3) flowing through a stream tube
The radius of the tube at point 1 is 2 m, whereas the radius of the tube at point 2 is 2.03 m.
The flow enters the tube at point 1 at V_1= 34 m/s

What is the velocity at point 2, V_2?

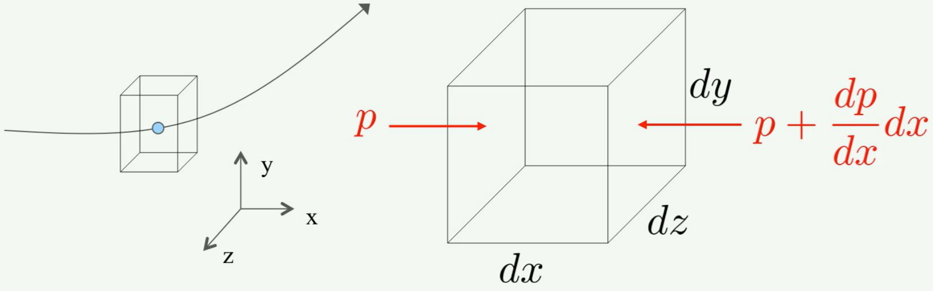
Fundamental Equations

Euler Equation

Momentum before = Momentum after
(Momentum is conserved)

Newton's second law: $F = m \cdot a$

Apply it to a flowing gas:



Fundamental Equations

Euler Equation

In reality **3 forces** act on the element:

- Pressure force
- Friction force
- Gravity force

In the derivation, we neglect:

- The gravity force (small)
- The viscosity → no friction forces

Fundamental Equations

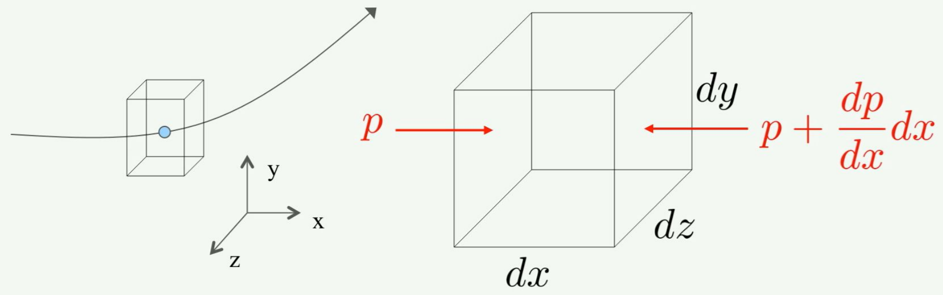
Euler Equation

$$\boxed{dp = -\rho V\, dV}$$

Gravity forces neglected
Viscosity neglected
Steady flow

Newton's second law: $F = m \cdot a$

Apply it to a flowing gas:



Fundamental Equations

Bernoulli Equation

Integrating Euler equation

Incompressible flow!

$$\boxed{dp = -\rho V \, dV} \longrightarrow \boxed{p + \frac{1}{2}\rho V^2 = \text{Constant along a streamline}}$$

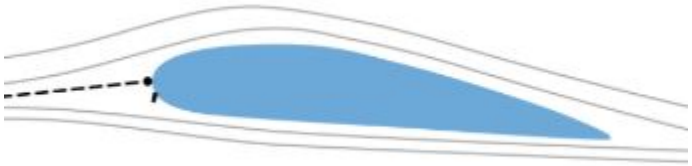
Fundamental Equations

Bernoulli Equation

That is why at the stagnation point (V = 0), pressure = total pressure

$$p + \frac{1}{2}\rho V^2 = p_t$$

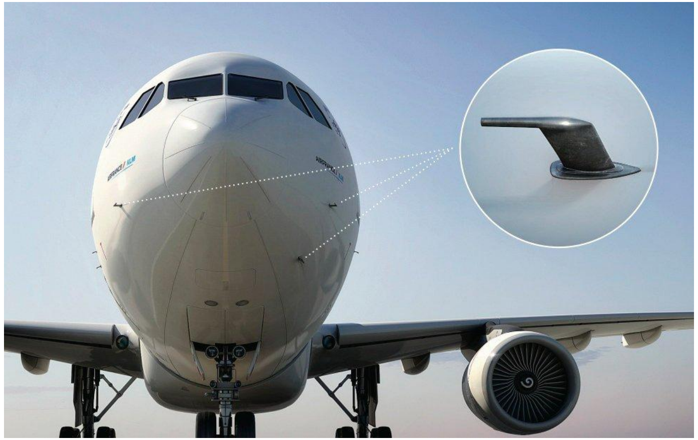
Static pressure + Dynamic pressure = Total pressure

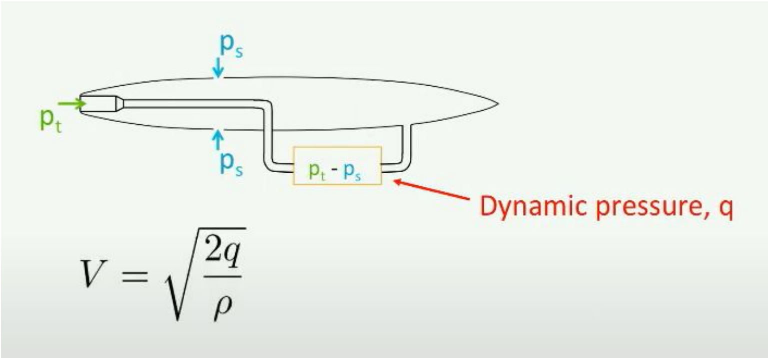


Fundamental Equations

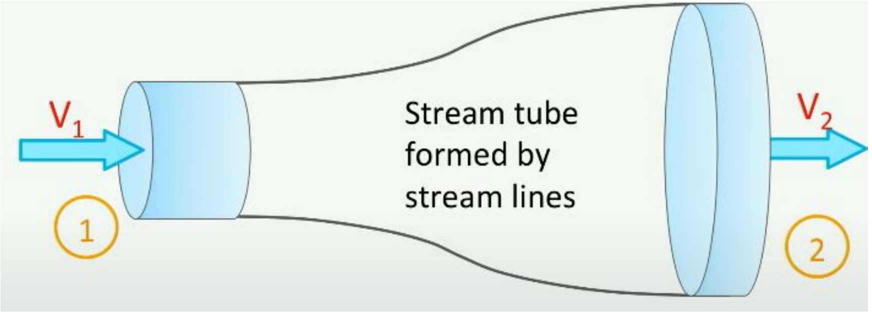
Bernoulli Equation

Why is it useful? Pitot + static port/ Pitotstatic tubes are used to find the flight speed





Fundamental Equations



Bernoulli

In the image you can see water (ρ = 1000 kg/m^3) flowing through a stream tube
The radius of the tube at point 1 is 2 m, whereas the radius of the tube at point 2 is 2.03 m.
The flow enters the tube at point 1 at V_1= 34 m/s

What is the velocity at point 2, V_2?

If the pressure at the exit, p_2, is 1 atm, what is the pressure at the inlet, p_1?

Fundamental Equations

Bernoulli

An aircraft flies at an altitude of 3000 m with a speed of 330 km/hr.

Calculate the pressure in the stagnation point of the wing.

At 3000 m the static pressure p is 70121 N/m^2 , the temperature T=268.67K and the density is 0.90926 kg/m^3

Exercises

We derived the Euler equation dp = -rho*VdV from Newton's 2nd law (Force = mass
*
acceleration) applied on a flow particle.

- What assumptions were made to come to Euler's equation?
- Use the Euler equation to derive Bernoulli's equation. Also here mention the assumptions you make.

Exercises

An airfoil was tested in a subsonic wind tunnel at a flow speed of 60 m/s at standard sea level conditions. The highest speed on the airfoil upper surface at a certain angle of attack occurred in point A and was 90 m/s.

Calculate the pressure at point A

Exercises

An aircraft is flying at 7000 m, where the temperature is -30.5ºC, and the pressure is 41060.35 Pa

The pitot tube measure a pressure p_t of 71,000 Pa

What is the velocity of the aircraft?